



2491 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Cycle: 1, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	GC-RC1-MIRI-AB	MIRI Medium Resolution Spectroscopy	(1) GC-RC1-MIRI
	2	GC-BG-MIRI-AB	MIRI Medium Resolution Spectroscopy	(3) MIRI-GC-BCKGR-1
	3	GC-RC1-MIRI-C	MIRI Medium Resolution Spectroscopy	(1) GC-RC1-MIRI
	4	GC-BG-MIRI-C	MIRI Medium Resolution Spectroscopy	(3) MIRI-GC-BCKGR-1
	5	GC-RC2-MIRI-ABC	MIRI Medium Resolution Spectroscopy	(2) GC-RC2-MIRI
	6	GC-BG-MIRI-ABC	MIRI Medium Resolution Spectroscopy	(3) MIRI-GC-BCKGR-1
Observation Folder				
	7	GC-RC1-MIRI-AB	MIRI Medium Resolution Spectroscopy	(1) GC-RC1-MIRI
	8	GC-BG-MIRI-AB	MIRI Medium Resolution Spectroscopy	(3) MIRI-GC-BCKGR-1
	9	GC-RC1-MIRI-C	MIRI Medium Resolution Spectroscopy	(1) GC-RC1-MIRI

Folder	Observation	Label	Observing Template	Science Target
	13	GC-RC1-MIRI-C	MIRI Medium Resolution Spectroscopy	(1) GC-RC1-MIRI
	14	GC-BG-MIRI-C	MIRI Medium Resolution Spectroscopy	(3) MIRI-GC-BCKGR-1
	23	GC-RC1-MIRI-C - Repeat obs 13	MIRI Medium Resolution Spectroscopy	(1) GC-RC1-MIRI
	24	GC-BG-MIRI-C - Repeat obs 14	MIRI Medium Resolution Spectroscopy	(3) MIRI-GC-BCKGR-1
	10	GC-BG-MIRI-C	MIRI Medium Resolution Spectroscopy	(3) MIRI-GC-BCKGR-1
	11	GC-RC2-MIRI-ABC	MIRI Medium Resolution Spectroscopy	(2) GC-RC2-MIRI
	12	GC-BG-MIRI-ABC	MIRI Medium Resolution Spectroscopy	(3) MIRI-GC-BCKGR-1

ABSTRACT

Despite the need for very high gas densities to overcome the tidal field of the supermassive black hole (Sgr A*) and induce gravitational collapse, there is increasing evidence for star formation taking place at the heart of the Milky Way. With the superb capabilities of the JWST MIRI MRS integral field spectrograph, we will finally be able to test the hypothesis of low-mass star formation at the Galactic Centre (GC). We propose to observe a group of faint mid-infrared sources in the nuclear star cluster to uncover their true nature. Their extreme infrared excess and spectral indices (based on long integration VLT L- and N-band data) and their association with photoevaporative protoplanetary disk-like counterparts in the radio, make them prime candidates for embedded low-mass young stars, located just about 0.5 pc in projection away from Sgr A*. Only MIRI MRS will be able to study and even characterise these sources in such unprecedented detail that was only possible so far for nearby star forming regions. Confirming the presence of low-mass young stars at the GC has implications on disk dispersion time-scales in dense clusters, which opens up the potential for planet formation in the tidally stressed environment of galactic nuclei.

OBSERVING DESCRIPTION

This proposal aims at using MIRI MRS to observe a group of protoplanetary disk candidates, located just half a parsec to the East from the central SMBH at the GC. The obtained MRS spectra will confirm their nature, and allow for a thorough characterisation of their properties.

The group of sources we intend to observe are arranged in two spatial groups, RC1 and RC2. Group RC1 will be observed in a 3x2 mosaic (with 10% overlap) to ensure that all sources are covered by the smaller FOV of channel 1. The second group RC2 will be observed in a 1x2 mosaic. This is sufficient to cover all sources of RC2 considering it is smaller than RC1.

We propose to use the full wavelength coverage of the MIRI MRS channel 1 to channel 4 with a 4-point dither pattern and FAST readout mode. We

JWST Proposal 2491 (Created: Tuesday, October 8, 2024, 9:00:10AM Eastern Standard Time) - Overview

will not use the simultaneous imaging due to saturation risk from the bright members of the nuclear cluster. We have also included a background pointing on a region far from the cluster to be able to measure the telescope's thermal emission. The background observations use the same detector configuration and a 2-point dither pattern and are linked in a non-interruptible sequence so that they are obtained immediately before and after one another.

We request a total telescope time of 11.61 hours (5.66 hours science time + overheads) to obtain reasonable SNRs in most channels for the faint compact sources (~5-15 mJy measured at 8.6 micron). The estimated exposure times were done in the ETC using multiple scenes and sources, corresponding to point and extended sources found in the FOV and a MIR spectrum of the Galactic Centre from ISO-SWS.

At the edge of the targeted FOV, there are a couple of MIR bright sources (part of the IRS 5 complex) that may cause partial saturation at the edges of the channel 4 detector, away from the science targets.

No target acquisition is required as the FOV and the targeted sources have known precise positions from radio observations.

Proposal 2491 - Targets - Low-mass Star Formation in the Inner Parsec of our Galaxy

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	GC-RC1-MIRI	RA: 17 45 41.1275 (266.4213646d) Dec: -29 00 25.03 (-29.00695d) Equinox: J2000		
	Comments: Category=Unidentified Description=[Infrared sources, Radio sources]				
	(2)	GC-RC2-MIRI	RA: 17 45 41.3601 (266.4223338d) Dec: -29 00 19.61 (-29.00545d) Equinox: J2000		
	Comments: Category=Unidentified Description=[Infrared sources, Radio sources]				
	(3)	MIRI-GC-BCKGR-1	RA: 17 45 45.8591 (266.4410796d) Dec: -29 01 29.90 (-29.02497d) Equinox: J2000		
	Comments: Category=Calibration Description=[Telescope/sky background]				

Proposal 2491 - Observation 1 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Observation	Proposal 2491, Observation 1: GC-RC1-MIRI-AB												Tue Oct 08 14:00:10 GMT 2024		
	Diagnostic Status: Warning														
	Observing Template: MIRI Medium Resolution Spectroscopy														
	Background Observations:[GC-BG-MIRI-AB (Obs 2)]														
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.														
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections				Miscellaneous				
	(1)	GC-RC1-MIRI		RA: 17 45 41.1275 (266.4213646d)											
				Dec: -29 00 25.03 (-29.00695d)											
				Equinox: J2000											
Acquisition	Comments: Category=Unidentified Description=[Infrared sources, Radio sources]														
	#							Target							
	1							NONE							
Template	AcqFilter		Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
	F560W		All MRS			NO			FULL			Allow Auto Reorder			
Mosaic	Rows	Columns		Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order			
	2	3		10.0		10.0		0.0		0.0		DEFAULT			
Dithers	#	Dither Type					Optimized For				Direction				
	1	4-Point					EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID		
	1	SHORT(A)	MRSLONG		FASTR1	70	1	1	Dither 1	4	4	777.011			
	1	SHORT(A)	MRSSHORT		FASTR1	70	1	1	Dither 1	4	4	777.011			
	2	MEDIUM(B)	MRSLONG		FASTR1	70	1	1	Dither 1	4	4	777.011			
	2	MEDIUM(B)	MRSSHORT		FASTR1	70	1	1	Dither 1	4	4	777.011			

Proposal 2491 - Observation 1 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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Proposal 2491 - Observation 2 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Observation	Proposal 2491, Observation 2: GC-BG-MIRI-AB												Tue Oct 08 14:00:10 GMT 2024	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [GC-RC1-MIRI-AB (Obs 1)]													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	MIRI-GC-BCKGR-1			RA: 17 45 45.8591 (266.4410796d) Dec: -29 01 29.90 (-29.02497d) Equinox: J2000									
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
		All MRS			NO			FULL			Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For			Direction				
	1	2-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SHORT(A)	MRSLONG		FASTR1	70	1	1	Dither 1	2	2	388.506		
	1	SHORT(A)	MRSSHORT		FASTR1	70	1	1	Dither 1	2	2	388.506		
	2	MEDIUM(B)	MRSLONG		FASTR1	70	1	1	Dither 1	2	2	388.506		
	2	MEDIUM(B)	MRSSHORT		FASTR1	70	1	1	Dither 1	2	2	388.506		

Proposal 2491 - Observation 2 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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Proposal 2491 - Observation 3 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Observation	Proposal 2491, Observation 3: GC-RC1-MIRI-C												Tue Oct 08 14:00:10 GMT 2024	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[GC-BG-MIRI-C (Obs 4)]													
Diagnostics	(Visit 3:1) Warning (Form): Data Excess over lower threshold													
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections				Miscellaneous			
	(1)	GC-RC1-MIRI		RA: 17 45 41.1275 (266.4213646d) Dec: -29 00 25.03 (-29.00695d) Equinox: J2000										
	Comments: Category=Unidentified Description=[Infrared sources, Radio sources]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
	F560W	All MRS			NO			FULL			Allow Auto Reorder			
Mosaic	Rows	Columns		Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order		
	2	3		10.0		10.0		0.0		0.0		DEFAULT		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	LONG(C)	MRSLONG		FASTR1	70	2	1	Dither 1	4	8	1565.123		
	1	LONG(C)	MRSSHORT		FASTR1	70	2	1	Dither 1	4	8	1565.123		

Proposal 2491 - Observation 3 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Special Requirements	Sequence Observations 3, 4, Non-interruptible
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Proposal 2491 - Observation 4 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Observation	Proposal 2491, Observation 4: GC-BG-MIRI-C												Tue Oct 08 14:00:10 GMT 2024	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [GC-RC1-MIRI-C (Obs 3)]													
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	MIRI-GC-BCKGR-1			RA: 17 45 45.8591 (266.4410796d) Dec: -29 01 29.90 (-29.02497d) Equinox: J2000									
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
		All MRS			NO			FULL			Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For			Direction				
	1	2-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	LONG(C)	MRSLONG		FASTR1	70	2	1	Dither 1	2	4	782.561		
	1	LONG(C)	MRSSHORT		FASTR1	70	2	1	Dither 1	2	4	782.561		

Proposal 2491 - Observation 4 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Special Requirements	Sequence Observations 3, 4, Non-interruptible
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Proposal 2491 - Observation 5 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Observation	Proposal 2491, Observation 5: GC-RC2-MIRI-ABC												Tue Oct 08 14:00:10 GMT 2024	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[GC-BG-MIRI-ABC (Obs 6)]													
Diagnostics	(Visit 5:1) Warning (Form): Data Excess over lower threshold													
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(2)	GC-RC2-MIRI	RA: 17 45 41.3601 (266.4223338d)											
			Dec: -29 00 19.61 (-29.00545d)											
			Equinox: J2000											
Acquisition	Comments:													
	Category=Unidentified													
Template	Description=[Infrared sources, Radio sources]													
	#	Target												
Mosaic	1	NONE												
	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
Dithers	F560W	All MRS			YES			FULL			Allow Auto Reorder			
	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order			
Spectral Elements	1	2	10.0		10.0		0.0		0.0		DEFAULT			
	#	Dither Type				Optimized For				Direction				
Spectral Elements	1	4-Point				EXTENDED SOURCE				NEGATIVE				
	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F560W	FASTR1	5	11	1	Dither 1	4	44	721.51		
	1	SHORT(A)	MRSLONG		FASTR1	70	1	1	Dither 1	4	4	777.011		
	1	SHORT(A)	MRSSHORT		FASTR1	70	1	1	Dither 1	4	4	777.011		
	2		IMAGER	F770W	FASTR1	5	11	1	Dither 1	4	44	721.51		
	2	MEDIUM(B)	MRSLONG		FASTR1	70	1	1	Dither 1	4	4	777.011		
	2	MEDIUM(B)	MRSSHORT		FASTR1	70	1	1	Dither 1	4	4	777.011		
	3		IMAGER	F1130W	FASTR1	5	19	1	Dither 1	4	76	1254.318		
	3	LONG(C)	MRSLONG		FASTR1	70	2	1	Dither 1	4	8	1565.123		
	3	LONG(C)	MRSSHORT		FASTR1	70	2	1	Dither 1	4	8	1565.123		

Proposal 2491 - Observation 5 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Special Requirements	Sequence Observations 5, 6, Non-interruptible
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Proposal 2491 - Observation 6 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Observation	Proposal 2491, Observation 6: GC-BG-MIRI-ABC												Tue Oct 08 14:00:10 GMT 2024	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [GC-RC2-MIRI-ABC (Obs 5)]													
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections				Miscellaneous		
	(3)	MIRI-GC-BCKGR-1	RA: 17 45 45.8591 (266.4410796d) Dec: -29 01 29.90 (-29.02497d) Equinox: J2000											
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				YES			FULL			Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For				Direction			
	1	2-Point					EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F560W	FASTR1	5	11	1	Dither 1	2	22	360.755		
	1	SHORT(A)	MRSLONG		FASTR1	70	1	1	Dither 1	2	2	388.506		
	1	SHORT(A)	MRSSHORT		FASTR1	70	1	1	Dither 1	2	2	388.506		
	2		IMAGER	F770W	FASTR1	5	11	1	Dither 1	2	22	360.755		
	2	MEDIUM(B)	MRSLONG		FASTR1	70	1	1	Dither 1	2	2	388.506		
	2	MEDIUM(B)	MRSSHORT		FASTR1	70	1	1	Dither 1	2	2	388.506		
	3		IMAGER	F1130W	FASTR1	5	19	1	Dither 1	2	38	627.159		
	3	LONG(C)	MRSLONG		FASTR1	70	2	1	Dither 1	2	4	782.561		
	3	LONG(C)	MRSSHORT		FASTR1	70	2	1	Dither 1	2	4	782.561		

Proposal 2491 - Observation 6 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Special Requirements	Sequence Observations 5, 6, Non-interruptible
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Proposal 2491 - Observation 7 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Observation	Proposal 2491, Observation 7: GC-RC1-MIRI-AB												Tue Oct 08 14:00:10 GMT 2024	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[GC-BG-MIRI-AB (Obs 8)]													
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections				Miscellaneous			
	(1)	GC-RC1-MIRI		RA: 17 45 41.1275 (266.4213646d) Dec: -29 00 25.03 (-29.00695d) Equinox: J2000										
	Comments: Category=Unidentified Description=[Infrared sources, Radio sources]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
	F560W	All MRS			NO			FULL			Allow Auto Reorder			
Mosaic	Rows	Columns		Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order		
	2	3		10.0		10.0		0.0		0.0		DEFAULT		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	MEDIUM(B)	MRSLONG		FASTR1	71	1	1	Dither 1	4	4	788.111		
	1	MEDIUM(B)	MRSSHORT		FASTR1	71	1	1	Dither 1	4	4	788.111		
	2	SHORT(A)	MRSLONG		FASTR1	71	1	1	Dither 1	4	4	788.111		
	2	SHORT(A)	MRSSHORT		FASTR1	71	1	1	Dither 1	4	4	788.111		

Proposal 2491 - Observation 7 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Special Requirements	Sequence Observations 7, 8, Non-interruptible
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Proposal 2491 - Observation 8 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Observation	Proposal 2491, Observation 8: GC-BG-MIRI-AB												Tue Oct 08 14:00:10 GMT 2024	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [GC-RC1-MIRI-AB (Obs 7)]													
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	MIRI-GC-BCKGR-1			RA: 17 45 45.8591 (266.4410796d) Dec: -29 01 29.90 (-29.02497d) Equinox: J2000									
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
		All MRS			NO			FULL			Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For				Direction			
	1	2-Point					EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	MEDIUM(B)	MRSLONG		FASTR1	71	1	1	Dither 1	2	2	394.056		
	1	MEDIUM(B)	MRSSHORT		FASTR1	71	1	1	Dither 1	2	2	394.056		
	2	SHORT(A)	MRSLONG		FASTR1	71	1	1	Dither 1	2	2	394.056		
	2	SHORT(A)	MRSSHORT		FASTR1	71	1	1	Dither 1	2	2	394.056		

Proposal 2491 - Observation 8 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Special Requirements	Sequence Observations 7, 8, Non-interruptible
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Proposal 2491 - Observation 9 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Observation	Proposal 2491, Observation 9: GC-RC1-MIRI-C												Tue Oct 08 14:00:10 GMT 2024	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[GC-BG-MIRI-C (Obs 10)]													
Diagnostics	(Visit 9:1) Warning (Form): Data Excess over lower threshold													
	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections				Miscellaneous			
	(1)	GC-RC1-MIRI		RA: 17 45 41.1275 (266.4213646d)										
				Dec: -29 00 25.03 (-29.00695d)										
				Equinox: J2000										
Acquisition	Comments:													
	Category=Unidentified													
	Description=[Infrared sources, Radio sources]													
	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
	F560W	All MRS			NO			FULL			Allow Auto Reorder			
Mosaic	Rows	Columns		Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order		
	2	3		10.0		10.0		0.0		0.0		DEFAULT		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	LONG(C)	MRSLONG		FASTR1	72	2	1	Dither 1	4	8	1609.523		
	1	LONG(C)	MRSSHORT		FASTR1	72	2	1	Dither 1	4	8	1609.523		

Proposal 2491 - Observation 9 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Special Requirements	Sequence Observations 9, 10, Non-interruptible
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Proposal 2491 - Observation 13 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Observation	Proposal 2491, Observation 13: GC-RC1-MIRI-C												Tue Oct 08 14:00:10 GMT 2024	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[GC-BG-MIRI-C (Obs 14)]													
Diagnostics	(Visit 13:1) Warning (Form): Data Excess over lower threshold													
	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections				Miscellaneous			
	(1)	GC-RC1-MIRI		RA: 17 45 41.1275 (266.4213646d) Dec: -29 00 25.03 (-29.00695d) Equinox: J2000										
	Comments: Category=Unidentified Description=[Infrared sources, Radio sources]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
	F560W	All MRS			NO			FULL			Allow Auto Reorder			
Mosaic	Rows	Columns		Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order		
	2	3		10.0		10.0		0.0		0.0		DEFAULT		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	LONG(C)	MRSLONG		FASTR1	72	2	1	Dither 1	4	8	1609.523		
	1	LONG(C)	MRSSHORT		FASTR1	72	2	1	Dither 1	4	8	1609.523		

Proposal 2491 - Observation 13 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Special Requirements	Sequence Observations 13, 14, Non-interruptible
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Proposal 2491 - Observation 14 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Observation	Proposal 2491, Observation 14: GC-BG-MIRI-C												Tue Oct 08 14:00:10 GMT 2024	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [GC-RC1-MIRI-C (Obs 13)]													
Diagnostics	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	MIRI-GC-BCKGR-1			RA: 17 45 45.8591 (266.4410796d) Dec: -29 01 29.90 (-29.02497d) Equinox: J2000									
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
		All MRS			NO			FULL			Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For			Direction				
	1	2-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	LONG(C)	MRSLONG		FASTR1	72	2	1	Dither 1	2	4	804.762		
	1	LONG(C)	MRSSHORT		FASTR1	72	2	1	Dither 1	2	4	804.762		

Proposal 2491 - Observation 14 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Special Requirements	Sequence Observations 13, 14, Non-interruptible
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Proposal 2491 - Observation 23 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Observation	Proposal 2491, Observation 23: GC-RC1-MIRI-C - Repeat obs 13												Tue Oct 08 14:00:10 GMT 2024		
	Diagnostic Status: Warning														
	Observing Template: MIRI Medium Resolution Spectroscopy														
	Background Observations:[GC-BG-MIRI-C - Repeat obs 14 (Obs 24)]														
Diagnostics	(Visit 23:1) Warning (Form): Data Excess over lower threshold														
	(Visit 23:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.														
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections				Miscellaneous				
	(1)	GC-RC1-MIRI		RA: 17 45 41.1275 (266.4213646d)											
				Dec: -29 00 25.03 (-29.00695d)											
				Equinox: J2000											
Acquisition	Comments: Category=Unidentified Description=[Infrared sources, Radio sources]														
	#							Target							
	1							NONE							
Template	AcqFilter		Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
	F560W		All MRS			NO			FULL			Allow Auto Reorder			
Mosaic	Rows	Columns		Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order			
	2	3		10.0		10.0		0.0		0.0		DEFAULT			
Dithers	#	Dither Type					Optimized For				Direction				
	1	4-Point					EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID		
	1	LONG(C)	MRSLONG		FASTR1	72	2	1	Dither 1	4	8	1609.523			
	1	LONG(C)	MRSSHORT		FASTR1	72	2	1	Dither 1	4	8	1609.523			

Proposal 2491 - Observation 23 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Special Requirements	Sequence Observations 23, 24, Non-interruptible
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Proposal 2491 - Observation 24 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Observation	Proposal 2491, Observation 24: GC-BG-MIRI-C - Repeat obs 14												Tue Oct 08 14:00:10 GMT 2024	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [GC-RC1-MIRI-C - Repeat obs 13 (Obs 23)]													
Diagnostics	(Visit 24:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	MIRI-GC-BCKGR-1			RA: 17 45 45.8591 (266.4410796d) Dec: -29 01 29.90 (-29.02497d) Equinox: J2000									
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
		All MRS			NO			FULL			Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For				Direction			
	1	2-Point					EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	LONG(C)	MRSLONG		FASTR1	72	2	1	Dither 1	2	4	804.762		
	1	LONG(C)	MRSSHORT		FASTR1	72	2	1	Dither 1	2	4	804.762		

Proposal 2491 - Observation 24 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Special Requirements	Sequence Observations 23, 24, Non-interruptible
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Proposal 2491 - Observation 10 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Observation	Proposal 2491, Observation 10: GC-BG-MIRI-C												Tue Oct 08 14:00:10 GMT 2024	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [GC-RC1-MIRI-C (Obs 9)]													
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	MIRI-GC-BCKGR-1			RA: 17 45 45.8591 (266.4410796d) Dec: -29 01 29.90 (-29.02497d) Equinox: J2000									
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
		All MRS			NO			FULL			Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For			Direction				
	1	2-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	LONG(C)	MRSLONG		FASTR1	72	2	1	Dither 1	2	4	804.762		
	1	LONG(C)	MRSSHORT		FASTR1	72	2	1	Dither 1	2	4	804.762		

Proposal 2491 - Observation 10 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Special Requirements	Sequence Observations 9, 10, Non-interruptible
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Proposal 2491 - Observation 11 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Observation	Proposal 2491, Observation 11: GC-RC2-MIRI-ABC												Tue Oct 08 14:00:10 GMT 2024	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[GC-BG-MIRI-ABC (Obs 12)]													
Diagnostics	(Visit 11:1) Warning (Form): Data Excess over lower threshold													
	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(2)	GC-RC2-MIRI	RA: 17 45 41.3601 (266.4223338d) Dec: -29 00 19.61 (-29.00545d) Equinox: J2000											
	Comments: Category=Unidentified Description=[Infrared sources, Radio sources]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
	F560W	All MRS				YES				FULL		Allow Auto Reorder		
Mosaic	Rows	Columns	Row Overlap %			Column Overlap %			Row shift (deg)		Column shift (deg)		Tile Order	
	1	2	10.0			10.0			0.0		0.0		DEFAULT	
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/E xp	Exposures/Dit h	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1130W	FASTR1	6	20	1	Dither 1	4	80	1542.922		
	1	LONG(C)	MRSLONG		FASTR1	72	2	1	Dither 1	4	8	1609.523		
	1	LONG(C)	MRSSHORT		FASTR1	72	2	1	Dither 1	4	8	1609.523		
	2		IMAGER	F770W	FASTR1	6	10	1	Dither 1	4	40	765.911		
	2	MEDIUM(B)	MRSLONG		FASTR1	71	1	1	Dither 1	4	4	788.111		
	2	MEDIUM(B)	MRSSHORT		FASTR1	71	1	1	Dither 1	4	4	788.111		
	3		IMAGER	F560W	FASTR1	6	10	1	Dither 1	4	40	765.911		
	3	SHORT(A)	MRSLONG		FASTR1	71	1	1	Dither 1	4	4	788.111		
	3	SHORT(A)	MRSSHORT		FASTR1	71	1	1	Dither 1	4	4	788.111		

Proposal 2491 - Observation 11 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Special Requirements	Sequence Observations 11, 12, Non-interruptible
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Proposal 2491 - Observation 12 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Observation	Proposal 2491, Observation 12: GC-BG-MIRI-ABC												Tue Oct 08 14:00:10 GMT 2024	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [GC-RC2-MIRI-ABC (Obs 11)]													
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(3)	MIRI-GC-BCKGR-1	RA: 17 45 45.8591 (266.4410796d) Dec: -29 01 29.90 (-29.02497d) Equinox: J2000											
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
		All MRS				YES				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	2-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1130W	FASTR1	6	20	1	Dither 1	2	40	771.461		
	1	LONG(C)	MRSLONG		FASTR1	72	2	1	Dither 1	2	4	804.762		
	1	LONG(C)	MRSSHORT		FASTR1	72	2	1	Dither 1	2	4	804.762		
	2		IMAGER	F770W	FASTR1	6	10	1	Dither 1	2	20	382.956		
	2	MEDIUM(B)	MRSLONG		FASTR1	71	1	1	Dither 1	2	2	394.056		
	2	MEDIUM(B)	MRSSHORT		FASTR1	71	1	1	Dither 1	2	2	394.056		
	3		IMAGER	F560W	FASTR1	6	10	1	Dither 1	2	20	382.956		
	3	SHORT(A)	MRSLONG		FASTR1	71	1	1	Dither 1	2	2	394.056		
	3	SHORT(A)	MRSSHORT		FASTR1	71	1	1	Dither 1	2	2	394.056		

Proposal 2491 - Observation 12 - Low-mass Star Formation in the Inner Parsec of our Galaxy

Special Requirements	Sequence Observations 11, 12, Non-interruptible
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